

Glossary of Key Terms

Convolutional Neural Networks for Beginners

AI Foundations Series — Book 4

This glossary collects the main terms and concepts from Book 4 — convolutional networks for computer vision — arranged alphabetically for quick reference. Each definition reflects how the term is used in the book itself: plain language first, with just enough precision to be useful when reviewing a chapter or returning to a companion script. Beginners can read it straight through; reviewers can jump to any entry.

1

1×1 Convolution

A convolution with a one-by-one kernel that mixes information across channels at a single pixel without looking at neighbors. It is used to cheaply shrink or grow the number of channels, and it is the trick at the heart of Inception modules.

A

Anchor Box

A predefined box shape that a detector uses as a starting guess. Rather than predicting boxes from nothing, the network adjusts these reference shapes, which makes it easier to detect objects of different sizes and aspect ratios.

C

Channel

One slice of a volume's depth - for a color image, the red, green, and blue layers are three channels. As data flows through a CNN, each filter adds a channel to the output, so deeper layers have many channels, each detecting a different pattern.

CIFAR-10

A standard dataset of small color images in ten everyday categories (airplane, cat, truck, and so on), used in the book as the first end-to-end CNN training project.

Content Cost

In neural style transfer, the measure of how much a generated image's high-level content differs from the content image. Minimizing it keeps the subject of the picture recognizable while the style is changed.

Convolution

The core operation of a CNN: a small grid of weights (a kernel) is slid across an image, and at each stop it multiplies and sums the pixels it covers to detect a pattern. Detecting that pattern anywhere it appears is the engine inside every CNN.

D

Data Augmentation

Expanding a training set by applying label-preserving transformations - flips, crops, rotations, color shifts - so the model sees more variety and overfits less. Especially valuable when labeled images are scarce.

E

Embedding (Face)

A compact list of numbers that a network produces to represent a face, so that two pictures of the same person sit close together and different people sit far apart. Comparing embeddings, not raw pixels, is what makes face recognition work.

F

Feature Extraction

The first stage of transfer learning: freeze a pretrained network's learned layers and use them as a fixed pattern detector, training only a small new classifier on top. Fast, and effective when data is limited.

Feature Map

The output produced when a filter is convolved across an input - a grid showing where in the image that filter's pattern was found. Stacking many feature maps gives a layer its depth.

Filter (Kernel)

The small grid of learnable weights that a convolution slides across the input to detect one pattern, such as an edge or a texture. A convolutional layer has many filters, each learning a different pattern.

Fine-Tuning

The second stage of transfer learning: after reusing a pretrained network's features, you unfreeze some of its layers and continue training them at a low learning rate so they adapt to the new task.

G

Gram Matrix

A compact summary of which features tend to appear together in an image, computed from a layer's feature maps. In style transfer it captures “style” - textures and colors - separately from content, in just a few lines of code.

I

Inception Module

A building block that runs several convolutions of different sizes (and pooling) in parallel and stacks their outputs, letting the network look at multiple scales at once. It relies on 1×1 convolutions to stay efficient.

Intersection over Union (IoU)

A score from 0 to 1 measuring how much two boxes (or pixel masks) overlap: the area of their intersection divided by the area of their union. It is the standard yardstick for detection and segmentation accuracy.

L

Localization

Predicting not just what is in an image but where, by outputting a bounding box around the object. A step beyond classification and a building block of detection.

N

Non-Max Suppression (NMS)

A cleanup step in object detection that removes duplicate, overlapping boxes for the same object, keeping only the highest-scoring one. It turns a messy pile of candidate boxes into clean final detections.

O

Object Detection

Finding and labeling multiple objects in one image, each with its own bounding box. It combines classification (what) with localization (where), and modern detectors like YOLO do it in a single pass.

One-Shot Learning

Recognizing something from a single example - for face recognition, identifying a person from just one photo on file. It motivates comparing embeddings rather than training a fresh classifier per person.

P

Padding

Adding a border of (usually zero) pixels around an image before convolving, so the output does not shrink and edge pixels are not under-used. “Same” padding keeps the output the same size as the input.

Parameter Sharing

The idea that the same filter weights are reused at every position in the image, so a pattern learned in one place is detected everywhere for free. It is the main reason CNNs need far fewer parameters than dense networks.

Pooling

A down-sampling step that shrinks a feature map by summarizing small regions - max pooling keeps the strongest response in each window. It reduces computation and makes features more robust to small shifts.

R

Receptive Field

The region of the original image that a particular neuron can “see.” It starts small in early layers and grows as layers stack, so deep neurons respond to large, complex parts of the image.

Residual Connection

A shortcut that adds a layer's input to its output, letting the network learn a small change (a residual) rather than a full transformation. It is what lets very deep networks (ResNets) train without their signal degrading.

ResNet (Residual Network)

A deep architecture built from residual connections, which made it possible to train networks hundreds of layers deep. A workhorse backbone for many vision tasks.

S

Semantic Segmentation

Labeling every pixel in an image with the class it belongs to, producing a mask rather than a box. It answers “which pixels are road, which are car?” and is evaluated with per-class IoU.

Siamese Network

Two identical copies of the same network that turn each of two inputs into an embedding, so the two can be compared. It is the structure behind face verification and one-shot recognition.

Sparsity of Connections

The property that each output of a convolution depends only on a small local patch of the input, not the whole image. Together with parameter sharing, it makes convolutions efficient.

Stride

How many pixels a filter jumps each time it moves across the image. A larger stride skips positions, producing a smaller output and reducing computation.

Style Cost

In neural style transfer, the measure of how much a generated image's textures and colors (captured by Gram matrices) differ from the style image. Minimizing it transfers the artistic style.

T

Transfer Learning

Reusing a network already trained on a large dataset as a starting point for a new task, via feature extraction and fine-tuning. It gives strong results with little data and is one of the most useful habits in applied vision.

Transpose Convolution

A convolution that makes a feature map larger instead of smaller, used to rebuild full-resolution output. It is the up-sampling step in segmentation networks like U-Net.

Triplet Loss

A training objective that takes an anchor image, a positive (same identity), and a negative (different identity), and pushes the anchor closer to the positive than to the negative by a margin. It is what teaches a face network to produce useful embeddings.

U

U-Net

A segmentation architecture that compresses an image to understand the big picture, then expands it back to full resolution, using skip connections to recover fine detail. Its U shape gives it its name.

V

Verification vs. Recognition

Two related face tasks: verification asks “are these two faces the same person?” (a yes/no), while recognition asks “who is this?” against a database of many people. Recognition is the harder, one-to-many version.

Y

YOLO (You Only Look Once)

A family of fast object detectors that predict all boxes and classes for an image in a single forward pass, rather than scanning regions one at a time. Its speed made real-time detection practical.

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